

TradeVoice: evidence of the N-ATLaS integration

NAIC 2026, Innovation and Enterprise track, PS2 Voice-First Access. Live app:

<https://tradevoice.duckdns.org/app>. Code: this repository, branch `claude/tradevoice-handoff-b7942v` (submission commit `c50a209`).

N-ATLaS is an initiative of the Federal Ministry of Communications, Innovation and Digital Economy, and powered by Awarri Technologies.

This document holds the evidence the form asks for:

- **Section 2:** API call samples, the exact requests the app sends to N-ATLaS.
- **Section 3:** the benchmark scripts and their results.
- **Section 4:** the automated test log.
- **Section 5:** how evaluators can reproduce a live run on the server.

The design and code map are in `TradeVoice-NATLAS-integration.pdf`.

1. What runs, and where

N-ATLaS model	What TradeVoice uses it for	Where it runs
NCAIR1/N-ATLaS (Llama-3 8B, official weights, BF16)	Turning the trader's words into a record (JSON); merging two hearings of one voice note; answering Yorùbá, Hausa and Igbo questions first	Modal, 1 x NVIDIA L4, vLLM, OpenAI-compatible API (served as <code>natlas</code>)
NCAIR1/Yoruba-ASR	Hearing Yorùbá voice notes	Modal (<code>deploy/modal_asr.py</code>)
NCAIR1/Hausa-ASR	Hearing Hausa voice notes	Modal
NCAIR1/Igbo-ASR	Hearing Igbo voice notes	Modal
NCAIR1/NigerianAccentedEnglish	Hearing English and Pidgin; the second hearing of every Yorùbá, Hausa and Igbo note	Modal

Where the app calls them:

- the language model: `src/llm.py chat()`, from `src/extract.py`, `src/hearing.py` and `src/agent.py`;
- the speech models: `src/asr.py _natlas_transcribe()`.
- **N-ATLaS is first for every record** (`models.insert(0, "natlas")` in `src/llm.py`). Backups answer only if it is down or out of time: NVIDIA models for words, Intron for voice.
- **Live talk:** Intron's streaming speech hears, for speed; N-ATLaS still writes every record.
- **Every answer is logged** with the model that gave it (no words, names, numbers or amounts), so the share N-ATLaS answered is measured, not estimated (`src/events.py`, `/team`,

scripts/validation_report.py).

- **Monitoring:**

- `src/natlas_watch.py` keeps both Modal apps awake in market hours and alerts the team when they stop answering.
- `/team` counts N-ATLaS active users against the licence's 1,000 a month and warns at 800.

2. API call samples (the exact requests, captured from the app's own code)

The samples below were produced by running the app's own functions (`extract._rec_call + llm.chat, asr._natlas_transcribe`) with the server address pointed at a recording stub. They are the request bodies the live server sends to our Modal apps. The key is sent as `Authorization: Bearer ...` and is never shown. Section 5 shows live answers.

2a. A record from the trader's words: `POST {NATLAS_URL}/chat/completions`

```
{
  "model": "natlas",
  "temperature": 0.1,
  "repetition_penalty": 1.12,
  "max_tokens": 250,
  "chat_template_kwargs": {"date_string": "10 Oct 2026"},
  "response_format": {"type": "json_schema", "json_schema": {"name": "answer",
"schema": {
  "type": "object", "required": ["type", "amount", "customer"],
  "properties": {"type": {"enum": ["sale", "credit_sale", "payment_received",
"expense",
                                "credit_purchase", "payment_made"]},
    "item": "...", "quantity": "...", "unit": "...", "amount": "...",
"each": "...",
    "customer": "...", "due_date": "...", "confidence": "...", "note":
"..."}}}}},
  "messages": [
    {"role": "system", "content": "You turn a Nigerian market trader's voice note
(English, Nigerian Pidgin, or mixed) into ONE bookkeeping record.\nToday is 2026-10-10
(Saturday).\nReturn ONLY a JSON object with these keys: ... (3,260 characters,
src/extract.py SYSTEM_PROMPT) ...\nNever invent an amount or a name that was not
said.\nPeople already in this trader's book: Iya Bisi, Mama Ngozi. If the note names
one of them (even misheard or misspelt), use that exact spelling."},
    {"role": "user", "content": "Mrs Adaeze took 3 rolls of lace, 4,500 each, she will
pay later"},
    {"role": "assistant", "content": "{\"type\": \"credit_sale\", \"item\": \"lace\",
\\\"quantity\\\": 3, \\\"unit\\\": \"roll\", \\\"amount\\\": 4500, \\\"each\\\": true, \\\"customer\\\":
\\\"Mrs Adaeze\\\", \\\"due_date\\\": null, \\\"confidence\\\": 0.9, \\\"note\\\": null}\"},
    "... 5 more worked examples (user, assistant): 2 in Pidgin, 1 each in Yorùbá, Hausa
and Igbo (src/extract.py SHOTS) ...",
    {"role": "user", "content": "Iya Bisi took 2 bags of rice for 60000, she will pay
Friday"}
  ]
}
```

What the code does with the answer (`src/extract.py`, tests in `eval/test_guards.py`):

- reads the JSON;
- keeps the amount only if it was said;

- multiplies "each" prices itself;
- keeps names spelled as in the trader's book;
- fills gaps from the offline rules;
- shows the trader a check card.

Nothing is saved before the trader says yes.

2b. Hearing a voice note: `POST {NATLAS_ASR_URL}/transcribe` (multipart form)

```
file = note.wav          (16 kHz mono after light preparation)
lang = yoruba            (yoruba | hausa | igbo | english: picks NCAIR1/Yoruba-ASR,
Hausa-ASR, Igbo-ASR, NigerianAccentedEnglish)
also = english           (Yorùbá, Hausa and Igbo notes are heard a second time by
NigerianAccentedEnglish)
prompt= Iya Bisi, rice  (the trader's own customer names and items, as hints)
prep = 1
```

2c. Merging the two hearings: `POST {NATLAS_URL}/chat/completions`

```
{"model": "natlas", "temperature": 0.1, "repetition_penalty": 1.12, "max_tokens": 120,
 "chat_template_kwargs": {"date_string": "10 Oct 2026"},
 "messages": [{"role": "user", "content": "Two speech models heard the same short voice
note from a Nigerian market trader who may mix\nYoruba and English.\nA (Yoruba model):
<words from NCAIR1/Yoruba-ASR>\nB (English model): <words from
NCAIR1/NigerianAccentedEnglish>\nCustomer names and items in the trader's book: Iya
Bisi, rice\nWrite the ONE sentence the trader most likely said, in the words they used
(keep Yoruba words in Yoruba and\nEnglish words in English). Use a name from the book
only when A or B clearly sounds like it. Never add a number,\nan amount or a name that
is not in A or B. Reply with the sentence only."}]}
```

An amount is kept only if one of the two hearings actually contains it (`src/hearing.py`).

3. Benchmark: what N-ATLaS adds (scripts and results)

Setup:

- **Models:** N-ATLaS and its base model, Meta-Llama-3-8B-Instruct.
- **Same conditions:** the same GPU (Modal L4, vLLM), the same prompt and worked examples, one model at a time, no backups.
- **Scoring:** a sentence counts as right only if the record's type, amount and customer are all right.

How to rerun it:

- **Base model:** deploy it with `NATLAS_MODEL_ID=meta-llama/Meta-Llama-3-8B-Instruct modal deploy deploy/modal_natlas.py`.
- **N-ATLaS alone:** `python eval/run_eval.py --cases eval/cases_1000.jsonl --llm natlas --raw --workers 8`.
- **Base model alone:** the same with `--url <base app>/v1 --shots all`.
- **Full app:** the same without `--raw`.

- **Where everything is:** results in `eval/results/` , the log of all runs in `docs/RESULTS.md` .

1,000 sentences (`eval/cases_1000.jsonl` , 200 per language, 2 Oct 2026):

	N-ATLaS alone	Llama-3-8B alone	Full app (N-ATLaS + code checks)
All	69% (66 to 71)	51% (47 to 54)	100%
English	85%	59%	100%
Pidgin	62%	57%	100%
Yorùbá	69%	60%	100%
Hausa	67%	27%	100%
Igbo	62%	52%	100%
Amounts said in Nigerian number words	62%	12%	100%
Wrong amount written	68	170	0
Median time (warm GPU)	5.0 s	5.3 s	5.1 s

Read honestly:

- **The model comparison is the result:** N-ATLaS is 18 points above its base model, and Hausa improves the most.
- **The full app's 100% is not proof:** our rules were built on similar template sentences, and the rules alone also score 100% on this set.
- **Fresh sentences:** on 100 sentences written later from research on how traders talk (`eval/cases_research.jsonl` , never used for tuning), the full app scored 69% before fixes.
- **Not settled:** on 210 other sentences (`eval/cases_fresh.jsonl`) the base model was ahead in Yorùbá and Pidgin.
- **No native-speaker check yet:** the Yorùbá, Hausa and Igbo test sentences were written by the team.
- **What will decide:** real traders' words, measured in the pilot.

N-ATLaS mistakes found by the benchmark and now guarded (tests in `eval/test_guards.py`):

- copying a worked example's price;
- taking the balance instead of the part paid;
- re-spelling customer names, which would split one customer in two.

4. Automated test log (run 10 Oct 2026, this commit)

`python eval/run_all.py` and `NODE_PATH=$(npm root -g) node eval/browser_test.cjs:`

✓ test_accounts.py	46/46
✓ test_agent.py	30/30
✓ test_ask.py	51/51
✓ test_askbook.py	31/31
✓ test_auth.py	38/38
✓ test_backup.py	16/16
✓ test_chat_smart.py	118/118
✓ test_clock.py	39/39
✓ test_converse.py	24/24
✓ test_corrections.py	35/35
✓ test_degraded.py	32/32
✓ test_demo_flow.py	25/25
✓ test_design.py	10/10
✓ test_extras.py	47/47
✓ test_guards.py	39/39
✓ test_hearing.py	47/47
✓ test_intron_tts.py	15/15
✓ test_language.py	9/9
✓ test_list.py	25/25
✓ test_live_intron.py	38/38
✓ test_natlas.py	42/42
✓ test_natlas_evidence.py	12/12
✓ test_preflight.py	12/12
✓ test_sms.py	36/36
✓ test_team.py	32/32
✓ test_telegram.py	42/42
✓ test_tools.py	47/47
✓ test_training.py	22/22
✓ test_validation_report.py	9/9
✓ test_whatsapp.py	19/19
✓ test_whatsapp_send.py	22/22
✓ test_wholesale.py	20/20

1030/1030 checks pass in 32 suites

80/80 browser checks pass (eval/browser_test.cjs, real Chromium)

What the suites cover:

- **Integration checks:**
 - `test_natlas.py`: N-ATLaS first, its settings, waking, resting when down, the early start in live talk.
 - `test_degraded.py`: the app keeps serving traders when N-ATLaS is down.
 - `test_hearing.py`: two hearings merged, an amount kept only if heard.
 - `test_guards.py`: the N-ATLaS mistakes above.
 - `test_natlas_evidence.py`: the live evidence run in section 5.
- **The browser test:** sign-up, voice, live talk, check card, save, reminders, Telegram, and the privacy notice, in a real browser.

5. Live run on the server (for the evaluators, 15 to 17 Oct)

One command on the live server calls N-ATLaS exactly as the app does and prints:

- what is served;

- five record calls, one per language, with N-ATLaS's JSON, the time, and the record after code checks;
- five speech round trips through the N-ATLaS speech models;
- the N-ATLaS share of the pilot's AI answers and voice notes.

The command fetches the evidence script from the branch, so the app keeps running the submitted code and nothing restarts:

```
sudo bash -c 'cd /opt/tradevoice/app && git fetch -q origin claude/tradevoice-handoff-b7942v && git show FETCH_HEAD:deploy/server/natlas_evidence.sh | bash'
```

Script: `scripts/natlas_evidence.py`, tested in `eval/test_natlas_evidence.py`. The output holds no keys, no names and no phone numbers.

The output of this command goes here. The team runs it on the live server and adds the output before submitting.

6. Licence and use

- TradeVoice uses N-ATLaS under the N-ATLaS Terms of Use v1.0. The attribution sentence appears on the website, in the app (Me > About), in the privacy notice, on the team dashboard, in the README and in every NAIC document.
- **Active users:** the licence allows up to 1,000 a month, and `/team` counts them. Beyond that, or before charging traders, we will agree commercial terms with Awarri and the Ministry.
- **N-ATLaS is never used to profile or score traders.** The lender report is plain code over the trader's own book, shared only by the trader.